ZAGAZIG NATIONAL UNIVERSITY

Faculty of Computers and Informatics

Artificial Intelligence and Data Science Program

SYSTEM ANALYSIS AND DESIGN

PROJECT DOCUMENTATION

CARECORE: SMART HEALTHCARE MANAGEMENT SYSTEM

ACADEMIC YEAR: 2025-2026

STUDENT NAME	SECTION
Yousef Ahmed Mohamed Roshdy Mohamed Elnady Elnaggar	6

1. *Planning & Feasibility*

Defining the system request, conducting the feasibility study (economic, technical, and organisational), and creating the Gantt chart.

2. *System Analysis*

Gathering requirements, defining functional and non-functional requirements, and drawing Data Flow Diagrams (DFD).

3. *Database Design*

Designing the core data structure and building the Entity-Relationship Diagram (ERD).

4. *System & UI Design*

Creating UML diagrams (Use Case, Class, Sequence), designing user interfaces (UI mockups), and software testing planning.

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Introduction & Project Idea

Introduction

Currently, many private clinics rely heavily on paper-based systems to manage patient records and daily appointments. This manual process causes several critical problems, such as long waiting times, lost medical files, and general disorganisation.

In modern healthcare, relying on paper is no longer efficient. Digital transformation is essential to solve these issues. Moving to a digital system helps to reduce human errors, save time, and provide a much better and more organised healthcare experience for both patients and medical staff.

Project Idea

CareCore is a cloud-based Smart Healthcare Management System designed as a Software as a Service (SaaS). The primary goal of this project is to convert traditional clinics into 100% paperless environments.

The system connects three main users: the patient, the secretary, and the doctor. It provides a mobile interface for patients to book their appointments easily from home. For the clinic staff, it offers a web dashboard allowing secretaries to manage the daily schedule smoothly, and provides doctors with a secure Electronic Health Record (EHR) system to write and save digital prescriptions. This ensures a fast, secure, and highly organised workflow.

System Request

Description	Element
Yousef Elnaggar (CareCore Lead Developer).	Project Sponsor
To transition private clinics from disorganised paper-based records to a fully digital, efficient, and secure cloud environment.	Business Need
The system must allow patients to book appointments online. It must also enable doctors to create Electronic Health Records (EHR) and send automated SMS reminders to patients.	Business Requirements
Decrease patient waiting times by up to 35%, eliminate the cost of paper supplies, and increase the daily capacity of the clinic through better time management.	Business Value
The prototype must be completed within a strict four-month academic timeframe. It must also ensure high data privacy for all medical records.	Special Issues or Constraints

Feasibility Analysis

(Technical, Economic, Organisational)

Feasibility Analysis

Before proceeding with the system design, a thorough feasibility analysis was conducted to ensure the CareCore project is practically viable and will be successfully adopted by the target users.

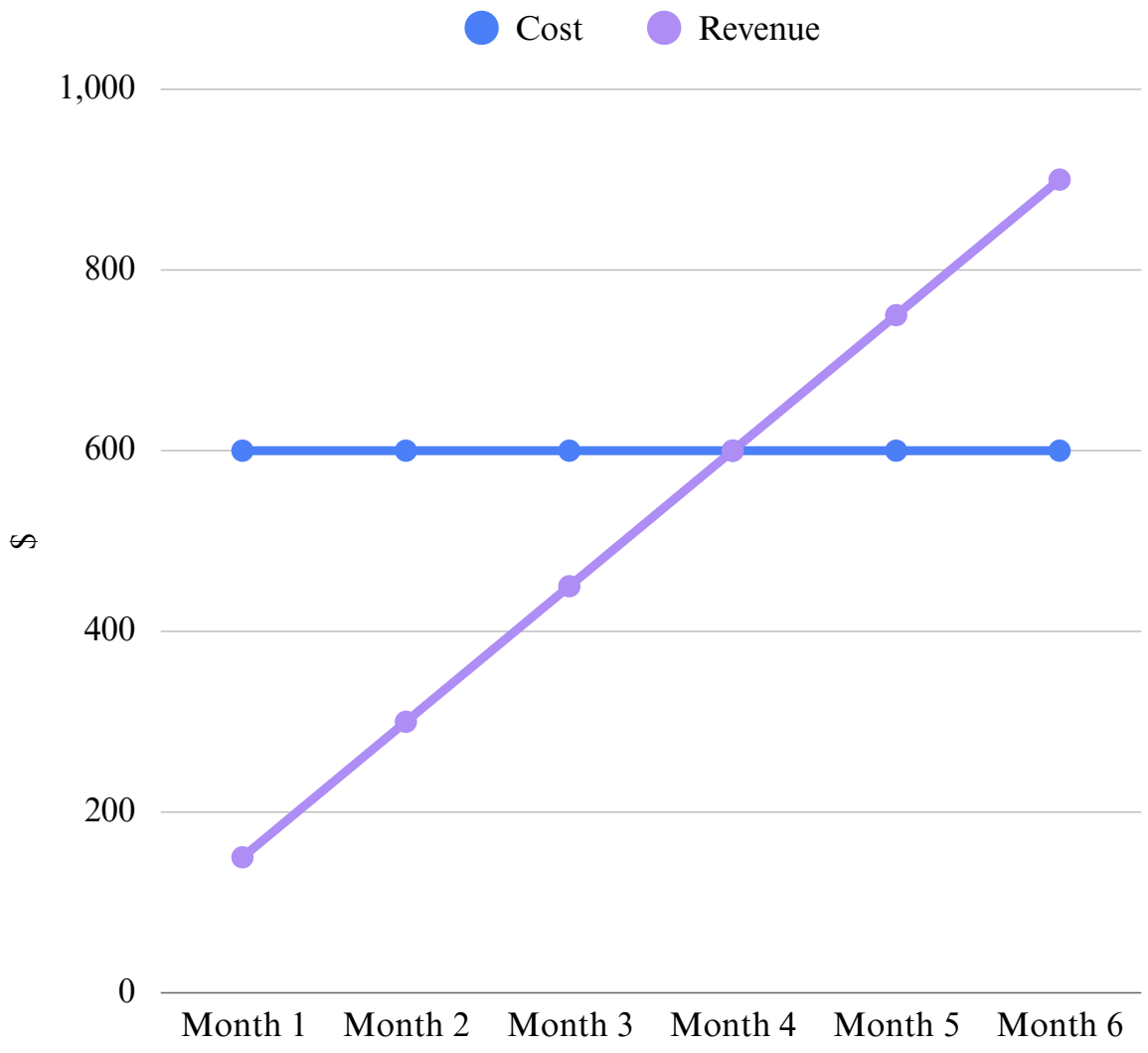
1. Technical Feasibility The technical feasibility of this project is highly positive.

- The system will be built utilising modern, well-documented programming languages and cloud infrastructures.
- As a second-year university student in the Artificial Intelligence and Data Science Program, I possess the required foundational skills in software engineering, database architecture, and user interface design to develop this prototype independently.
- The required hardware and cloud servers for hosting the database are easily accessible, highly reliable, and do not require complex initial setups.

2. Organisational Feasibility The organisational risk is very low, and the acceptance rate is expected to be extremely high.

- Clinic staff (doctors and secretaries) are currently overwhelmed and frustrated with lost paper files and disorganised daily queues. They will strongly welcome a digital system that simplifies their workflow.
- Patients increasingly prefer using mobile applications to manage their daily services, ensuring a rapid adoption rate for the CareCore mobile booking interface.

Economic Feasibility (Line Graph & ROI)



Economic Feasibility & ROI

The economic feasibility evaluates the financial viability of developing and operating the CareCore system. As an independent student project, the primary costs are minimal, focusing on cloud hosting and development time.

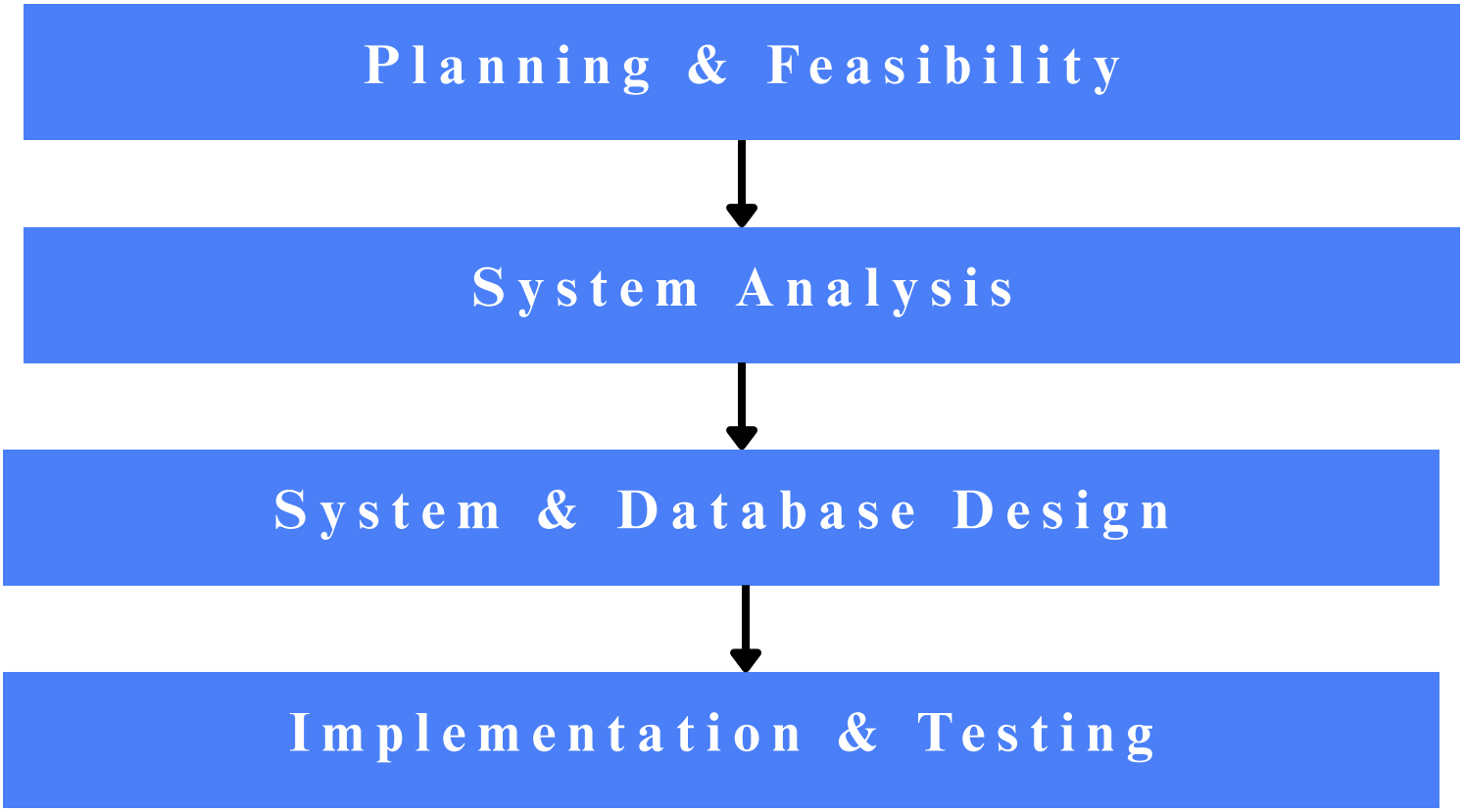
- Estimated Costs: The initial setup and server hosting costs are calculated at approximately $600 for the first operational cycle.
- Return on Investment (ROI): Clinics will subscribe to the platform as a Software as a Service (SaaS) for an estimated $150 per month.
- Break-even Analysis: As illustrated in the financial projection graph below, the steady accumulation of monthly subscriptions will cross the initial cost threshold precisely at the 4th month. This rapid break-even point proves the high economic efficiency of the project.

System Methodology (Waterfall Process)

System Methodology (Waterfall Process)

To ensure the highest engineering standards and a completely structured workflow, the Waterfall Methodology has been selected as the core software development life cycle (SDLC) model for the CareCore project.

This methodology was deliberately chosen because the requirements for a clinical management system are well-defined, strict, and stable from the beginning. It enforces a highly disciplined, sequential flow, which is crucial for maintaining focus and accuracy when developing a system independently.



As demonstrated in the process flow above, the Waterfall approach guarantees that every phase is thoroughly completed, reviewed, and fully documented before the next phase begins. This strict logical progression ensures that the final prototype will precisely match the initial system request without critical structural errors.

Project Plan & Gantt Chart

Project Plan (Work Breakdown Structure)

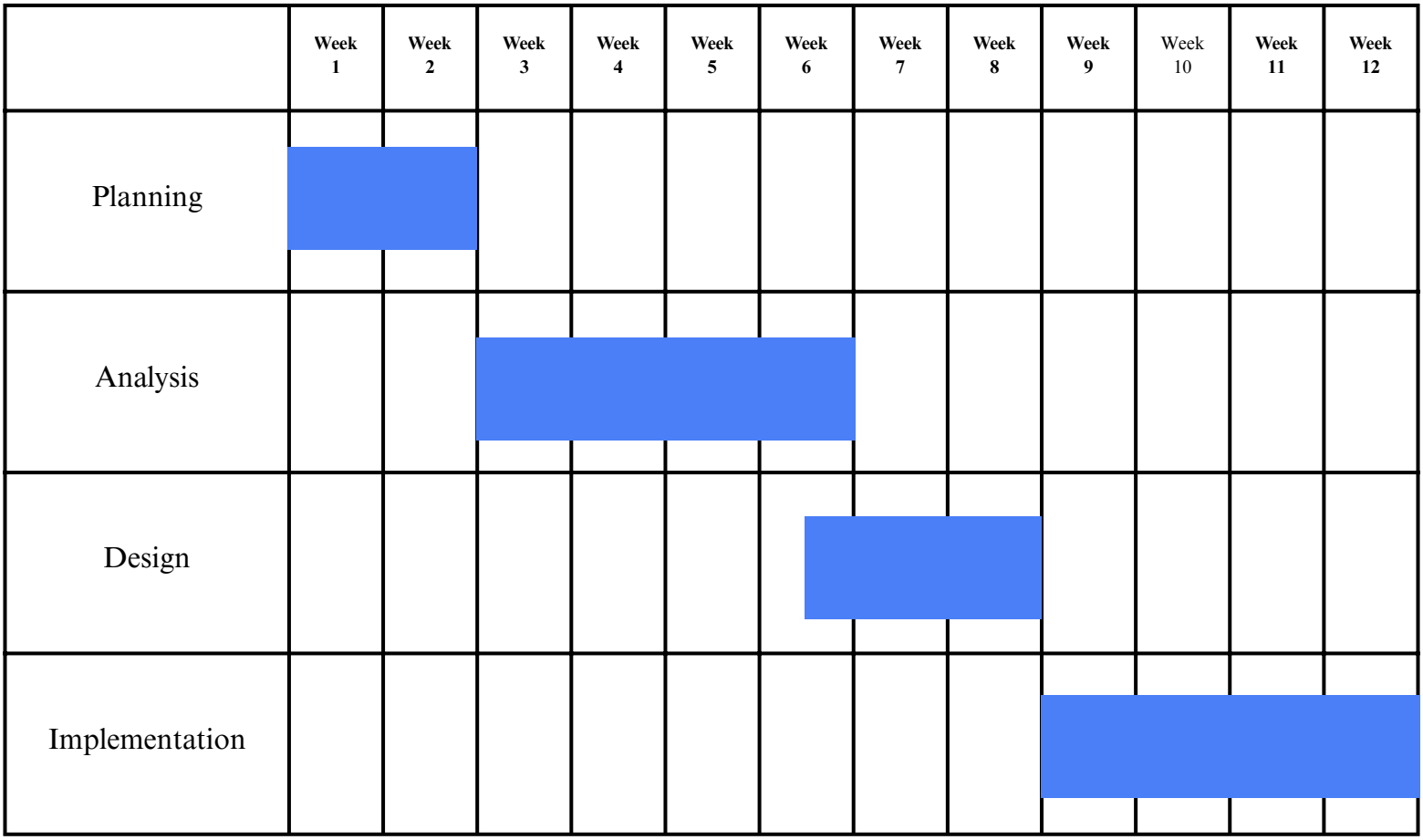
To effectively manage the development of the CareCore system as a solo developer, the project has been structured using a Work Breakdown Structure (WBS). This translates the Waterfall methodology into actionable engineering phases, ensuring all academic and technical deliverables are completed within a strict 84-day timeframe.

WBS ID	Project Phase & Core Tasks	Assigned To	Duration
1	**Planning:** Define the system request, analyse economic and technical feasibility, and finalise the project .scope	Yousef Elnaggar	Days 15
2	**Analysis:** Elicit requirements (interviews and surveys), document functional/nonfunction al constraints, and construct Data Flow .Diagrams (DFDs)	Yousef Elnaggar	Days 25
3	**Design:** Architect the database (ERD), map UML diagrams (Use Case, Class, Sequence), and create User .Interface (UI) mockups	Yousef Elnaggar	Days 20
4	**Implementation:** Finalise the technical documentation, review mathematical models (Alternative Matrix), and perform system .testing	Yousef Elnaggar	Days 24

Gantt Chart (Project Timeline)

Gantt Chart (Project Timeline)

The Gantt chart below provides a strict visual representation of the CareCore project timeline. It graphically maps the 84-day Work Breakdown Structure (WBS) across a 12-week academic schedule. This sequential representation proves that tasks do not overlap, strictly adhering to the selected Waterfall methodology and ensuring efficient time management by a single developer.



Business Model Canvas

(1. Key Partners).

- Cloud server providers (e.g., AWS, Google Cloud).
- Private clinic managers.

(2. Key Activities).

- System analysis and software architecture.
- Database management.
- Continuous security updates.

(3. Key Resources).

- Solo Software Developer (Yousef Elnaggar).
- Secure cloud database infrastructure.
- Cross-platform development frameworks.

(4. Value Propositions).

- 100% paperless clinical environment.
- Automated scheduling to eliminate patient waiting times.
- Centralised and highly secure Electronic Health Records (EHR).

(5. Customer Relationships).

- Automated self-service for patients (Mobile App).
- Technical documentation and support for medical staff.

(6. Channels).

- Mobile Application (Patient Access).
- Web Dashboard (Doctor and Secretary Access).

(7. Customer Segments).

- Forward-thinking private medical clinics.
- Patients seeking modern, digital healthcare services.

(8. Cost Structure).

- Initial setup and ongoing cloud hosting ($600 baseline).
- Developer time and architectural effort.
- Data encryption protocols (AES-256).

(9. Revenue Streams).

- Monthly Software as a Service (SaaS) subscription fee per clinic ($150/month).

Requirements Elicitation (Interview & Survey)

To ensure the CareCore system perfectly addresses real-world clinical bottlenecks, I conducted targeted requirement gathering sessions. As a solo developer, I utilised two primary techniques: direct interviews with medical staff and a digital survey distributed to patients.

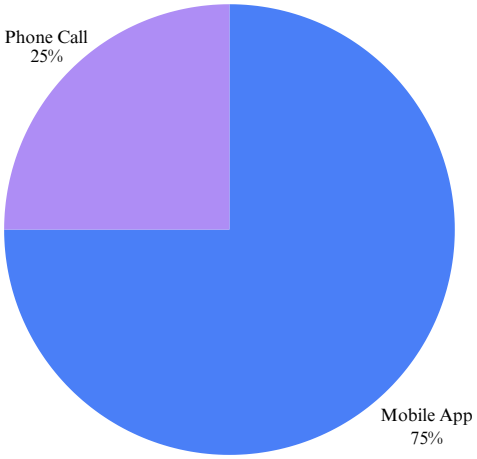
1. Interview with Clinic StaffTarget: Clinic Secretary and Lead Doctor. Objective: To understand the current operational workflow and pinpoint administrative inefficiencies.

- Question 1: What is the most time-consuming task in your daily workflow?
  - Answer: Searching for, organising, and retrieving paper-based patient records takes up to 40% of our daily administrative time. Files are frequently misplaced.
- Question 2: How do patients currently book their appointments?
  - Answer: Almost entirely via phone calls. This leads to constant interruptions, busy phone lines, and occasional human errors in the scheduling book.
- Question 3: What technological feature would assist the doctor the most?
  - Answer: A secure, digital platform to quickly write, save, and retrieve patient prescriptions (EHR) without dealing with messy paperwork.

2. Patient Survey Introduction To capture the user perspective, a digital survey was distributed to a sample of 100 patients. The objective was to identify their booking preferences and main frustrations with traditional clinics. The visual analysis of these results is presented in the following section.

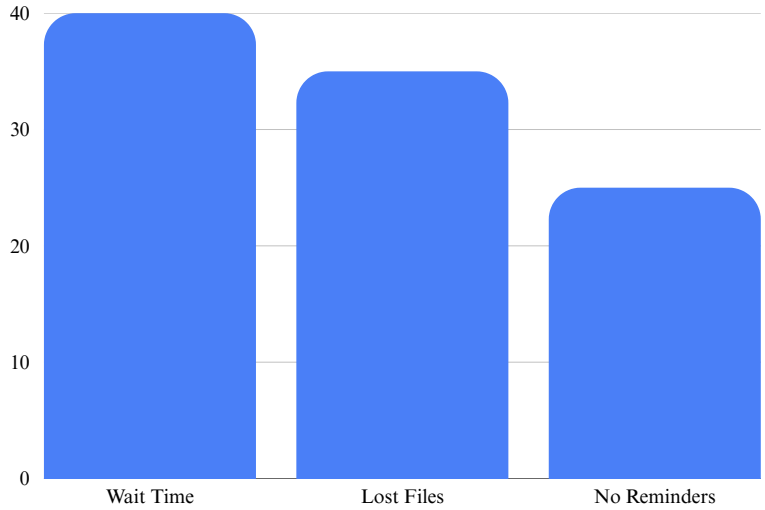
Survey Analysis (Pie & Bar Charts)

1. Patient Booking Preferences (Pie Chart) The survey data demonstrates a highly definitive demand for digital solutions. Exactly 75% of surveyed patients stated they prefer to book appointments via a mobile application, compared to only 25% who still prefer traditional phone calls. This statistically validates the necessity of developing the CareCore patient mobile interface.



2. Main Patient Complaints (Bar Graph) When patients were asked to identify their primary frustrations with current clinic operations, the data revealed three critical pain points:

- Wait Time (40%): Validates the need for an automated, strict scheduling system.
- Lost Files (35%): Directly justifies the implementation of a secure Electronic Health Record (EHR) module.
- No Reminders (25%): Highlights the absolute necessity of integrating an automated SMS notification feature.



System Functional Requirements

Functional requirements explicitly define the specific behaviours, actions, and features the CareCore system must execute to solve the clinical problems identified in the survey analysis.

ID	Module & User	Requirement Description
FR-01	Patient	**The system shall allow patients to register, securely log in, and manage their personal profiles .via the mobile app**
FR-02	Patient	**The system shall allow patients to view available clinic time slots and book an appointment online .in real-time**
FR-03	Patient	**The system shall allow patients to digitally cancel or reschedule their appointments up to 24 hours .in advance**
FR-04	Secretary	**The system shall enable the clinic secretary to view the daily schedule dashboard and manage .the patient queue to minimise wait times**
FR-05	Secretary	**The system shall empower the secretary to generate daily automated attendance and .financial summary reports**
FR-06	Doctor	**The system shall provide the doctor with strictly secure access to the patient's Electronic Health .Record (EHR) history**
FR-07	Doctor	**The system shall allow the doctor to create, save, and instantly retrieve digital medical .prescriptions**
FR-08	System	**The system shall automatically send an SMS reminder to patients 24 hours before their .scheduled visit to reduce no-shows**

System Nonfunctional Requirements

System Nonfunctional Requirements

Nonfunctional requirements specify the quality attributes, performance constraints, and strict security standards of the CareCore system. These criteria are essential to ensure the software is highly reliable, robust, and compliant with international data protection protocols.

ID	Category	Requirement Description
NFR-01	Performance	**The mobile application and web dashboard must render and load requested data within 2.0 seconds under standard network conditions to ensure .optimal user experience**
NFR-02	Security	**All sensitive patient medical records and authentication credentials must be strictly encrypted in the database .utilising the AES-256 standard**
NFR-03	Compliance	**The system architecture must adhere to global data privacy principles (aligning with European GDPR standards), ensuring strict Role-Based Access .Control (RBAC)**
NFR-04	Availability	**The cloud-hosted infrastructure shall guarantee an uptime of 99.9%, ensuring continuous access for medical staff .during critical clinic hours**
NFR-05	Usability	**The user interface (UI) must be highly intuitive, requiring no more than three screen taps for a patient to successfully .complete an appointment booking**
NFR-06	Scalability	**The backend database must be dynamically scalable to handle simultaneous read/write requests from multiple clinic branches without .performance degradation**

System Proposal (UML Overview)

Transitioning from the requirement elicitation phase, this section presents the technical proposal and architectural blueprint for the CareCore system. To translate the functional requirements into a structured, programmable format, Unified Modelling Language (UML) has been heavily utilised.
As a solo software developer, using UML diagrams is a critical engineering practice. It provides a standardised visual representation of the system's design, bridging the gap between theoretical business logic and actual code implementation. The following pages will detail the precise mechanics of the system through four core perspectives:

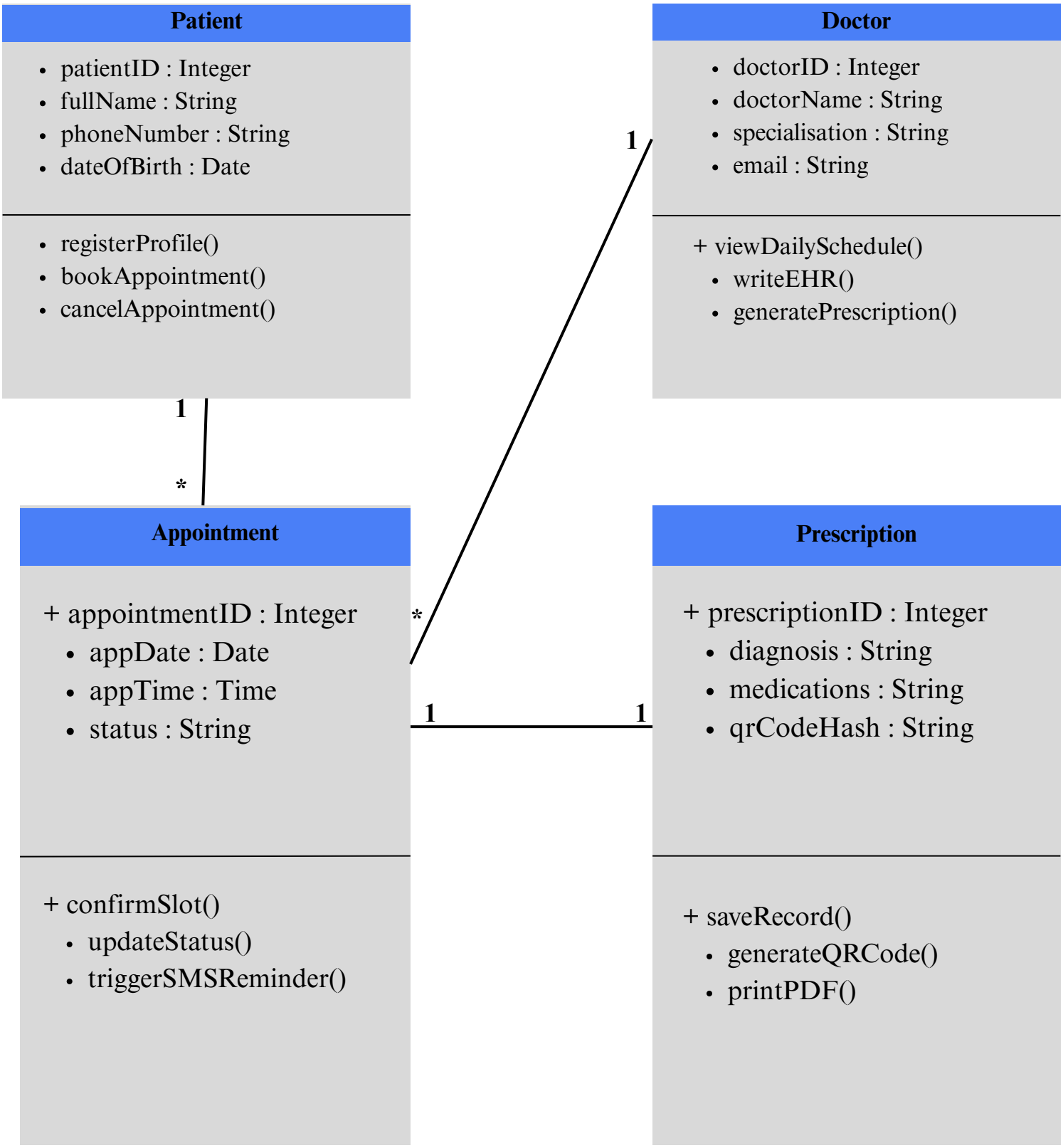
- 1. Structural Modelling (Class Diagram): Defines the static architecture, core entities, and their underlying relationships within the database.
- 2. Behavioural Interaction (Sequence Diagram): Maps the chronological flow of data and messages during critical operations, such as appointment booking.
- 3. Process Flow (Activity Diagram): Illustrates the dynamic decision-making paths and logical conditions within the software.
- 4. User Interaction (Usecase Figure): Provides a high-level summary of the exact permissions and actions assigned to each specific actor (Patient, Doctor, Secretary).

This meticulous architectural preparation guarantees that the coding phase will be highly efficient, significantly reducing the risk of structural errors or security vulnerabilities.

Class Diagram

Class Diagram

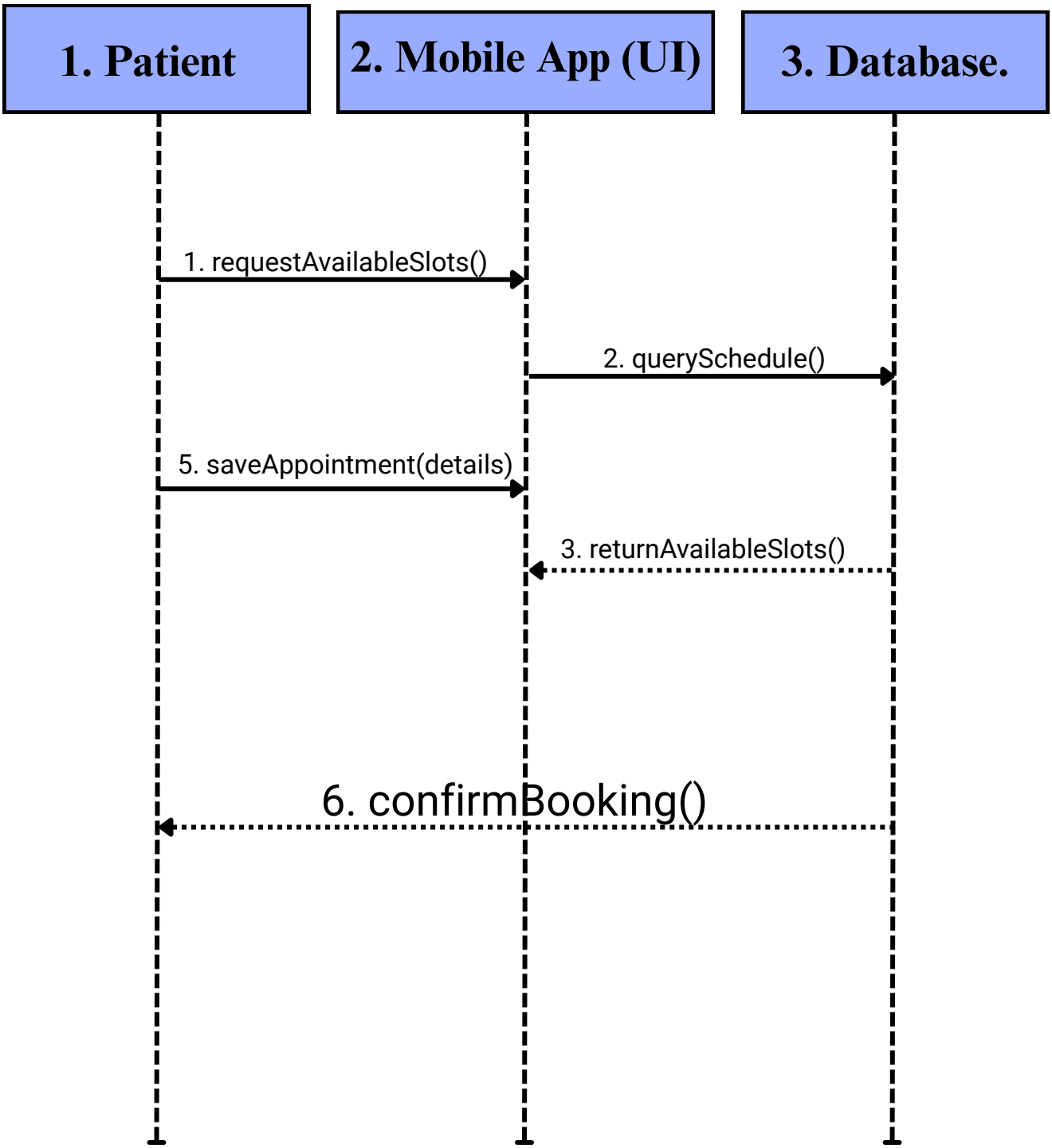
The Class Diagram establishes the static blueprint of the CareCore system. It meticulously defines the core classes, their internal attributes, operational methods, and structural multiplicities, serving as the direct foundation for the backend database schema.



Sequence Diagram

Sequence Diagram

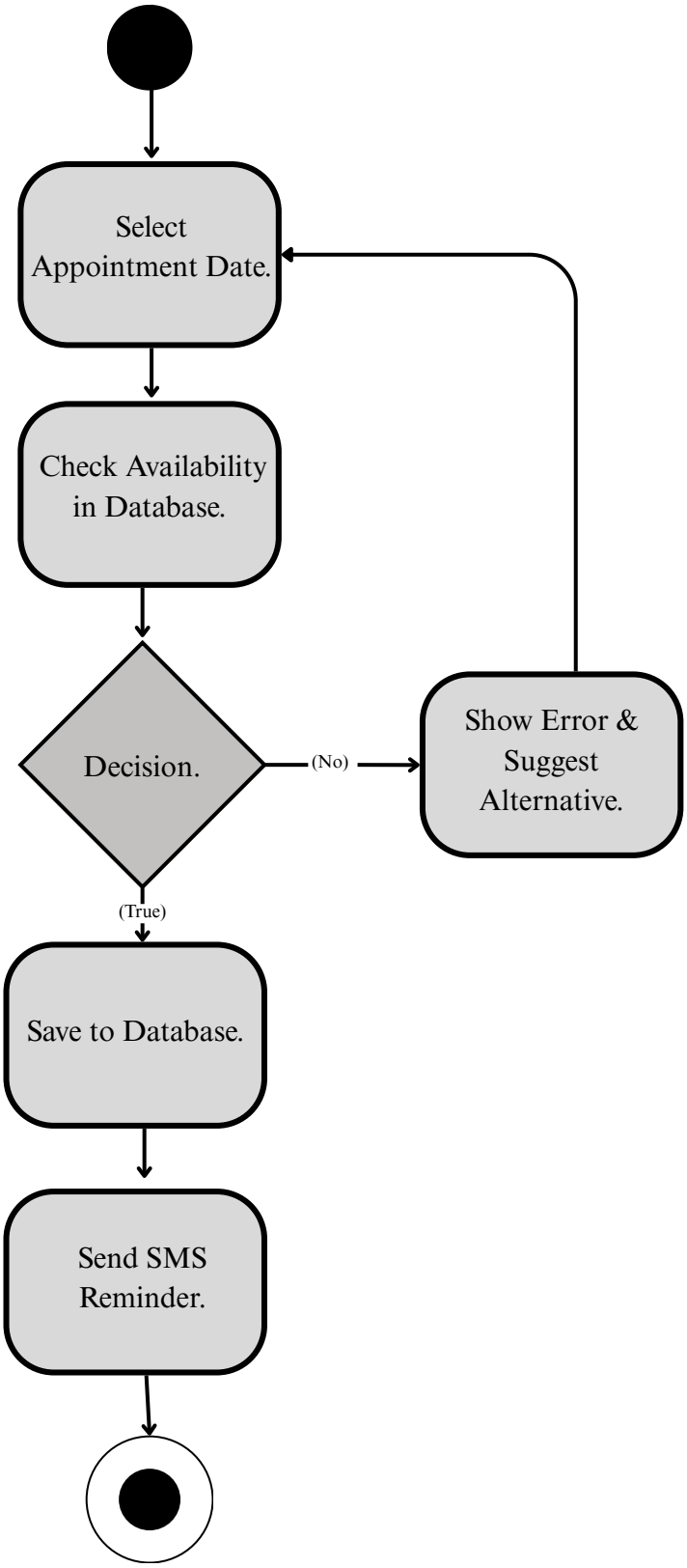
The Sequence Diagram details the chronological flow of messages and data exchanged between the system components during a specific operation. As a core feature of the CareCore platform, the diagram below maps out the precise sequence for the "Book Appointment" process, ensuring a secure and logical backend interaction.



Activity Diagram

Activity Diagram

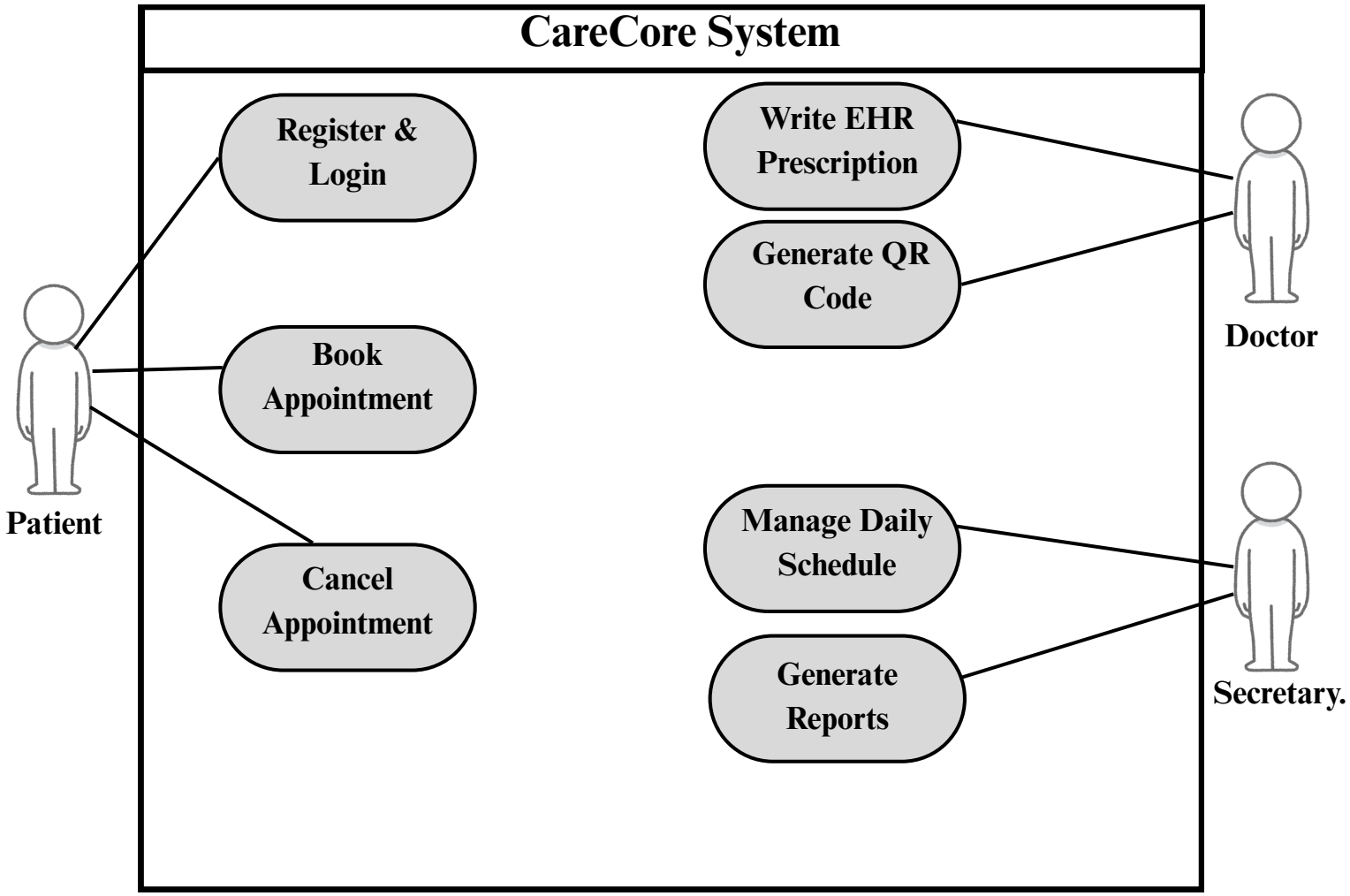
The Activity Diagram provides a visual flowchart of the dynamic decision-making pathways within the system. The following model illustrates the logical conditions the software evaluates when a patient requests a booking, specifically demonstrating how the system automatically handles schedule conflicts and prevents double-booking.



Usecase Figure

Usecase Figure

The Use Case Diagram maps the exact functional capabilities provided to each system actor. It serves as a visual summary of the System Functional Requirements (FRs) documented earlier. As a solo developer, constructing this diagram ensures that appropriate access control boundaries are established between the Patient, the Doctor, and the Secretary within the CareCore environment.



Use Case Tables (1)

Use Case Tables (1): Patient Booking Process

To provide a precise logical framework for the software implementation phase, the following table details the step-by-step sequential interaction for the most critical patient-facing use case: Booking an Appointment. This ensures every possible scenario, including system exceptions, is strictly handled.

Use Case Attribute	Detailed Description
Use Case Name	Book an Appointment
Primary Actor	Patient
Pre-conditions	The patient must be successfully registered and securely logged into the .mobile application
Main Success Course	The patient selects "Book New Appointment" from the .1 .dashboard The system retrieves and displays available doctors and .2 .open time slots .The patient selects a suitable date and time .3 ."The patient taps "Confirm Booking .4 The system saves the record to 2. Developer Profile .5 & Rolesthe database and schedules an automated .SMS reminder
Alternative Course	*Schedule Conflict:* If the selected time slot is simultaneously booked by another user, the system halts the save process, displays an error message, and .refreshes the available slots
Post-conditions	A new appointment entity is created in the database, and the daily clinic schedule is .automatically updated

Use Case Tables (2)

Use Case Tables (2): Doctor EHR Management

This table dissects the primary functional workflow for the medical staff. It outlines the strict protocol a doctor must follow to generate a secure Electronic Health Record (EHR) and a verifiable digital prescription, maintaining compliance with the defined nonfunctional security requirements.

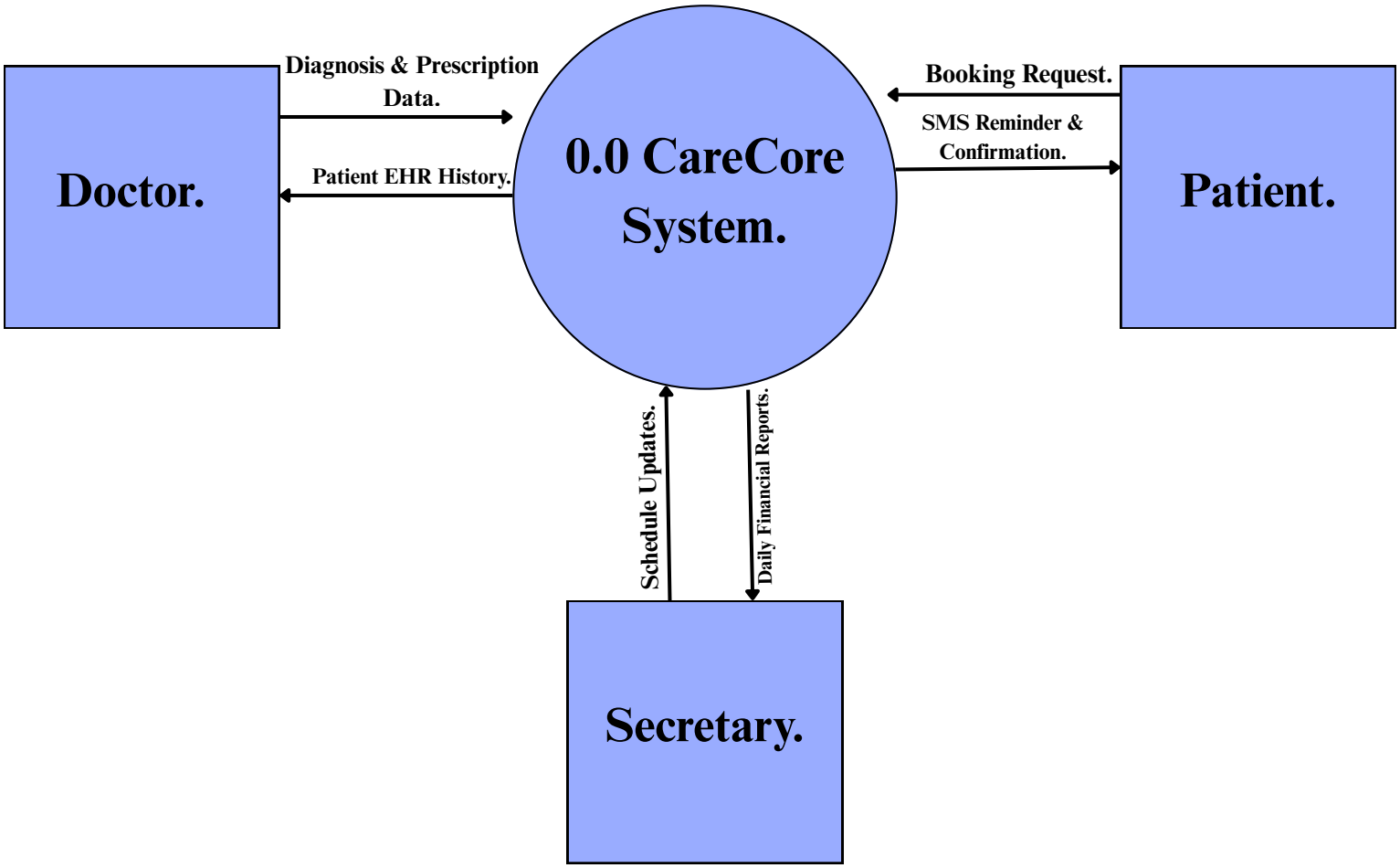
Use Case Attribute	Detailed Description
Use Case Name	Write EHR Prescription
Primary Actor	Doctor
Pre-conditions	The doctor must be securely logged in, and the patient's appointment status must be ."marked as "Active" or "Completed
Main Success Course	The doctor selects the patient's profile from the .1 .daily queue .The system opens a blank digital diagnostic form .2 The doctor inputs the medical diagnosis, required .3 .medications, and clinical notes ."The doctor clicks "Save & Generate Prescription .4 The system encrypts the record, generates a .5 unique QR Code, and saves it to the patient's .permanent history
Alternative Course	*Missing Data Validation:* If mandatory fields (e.g., medication name) are left blank, the system disables the save button and highlights the .missing fields in red
Post-conditions	The EHR database is securely updated, and a digital prescription is ready for printing or mobile .viewing

Context Diagram (DFD Level 0)

Context Diagram (DFD Level 0)

The Context Diagram, also referred to as Data Flow Diagram (DFD) Level 0, establishes the absolute system boundary of the CareCore platform. It deliberately treats the entire software architecture as a single, high-level process (Process 0.0) to clearly illustrate its external interactions.

As mapped below, the diagram highlights the exact data inputs and outputs flowing between the central system and its three primary external entities: the Patient, the Doctor, and the Secretary. This high-level abstraction is crucial before breaking the system down into detailed sub-processes in the subsequent Level 1 diagram.

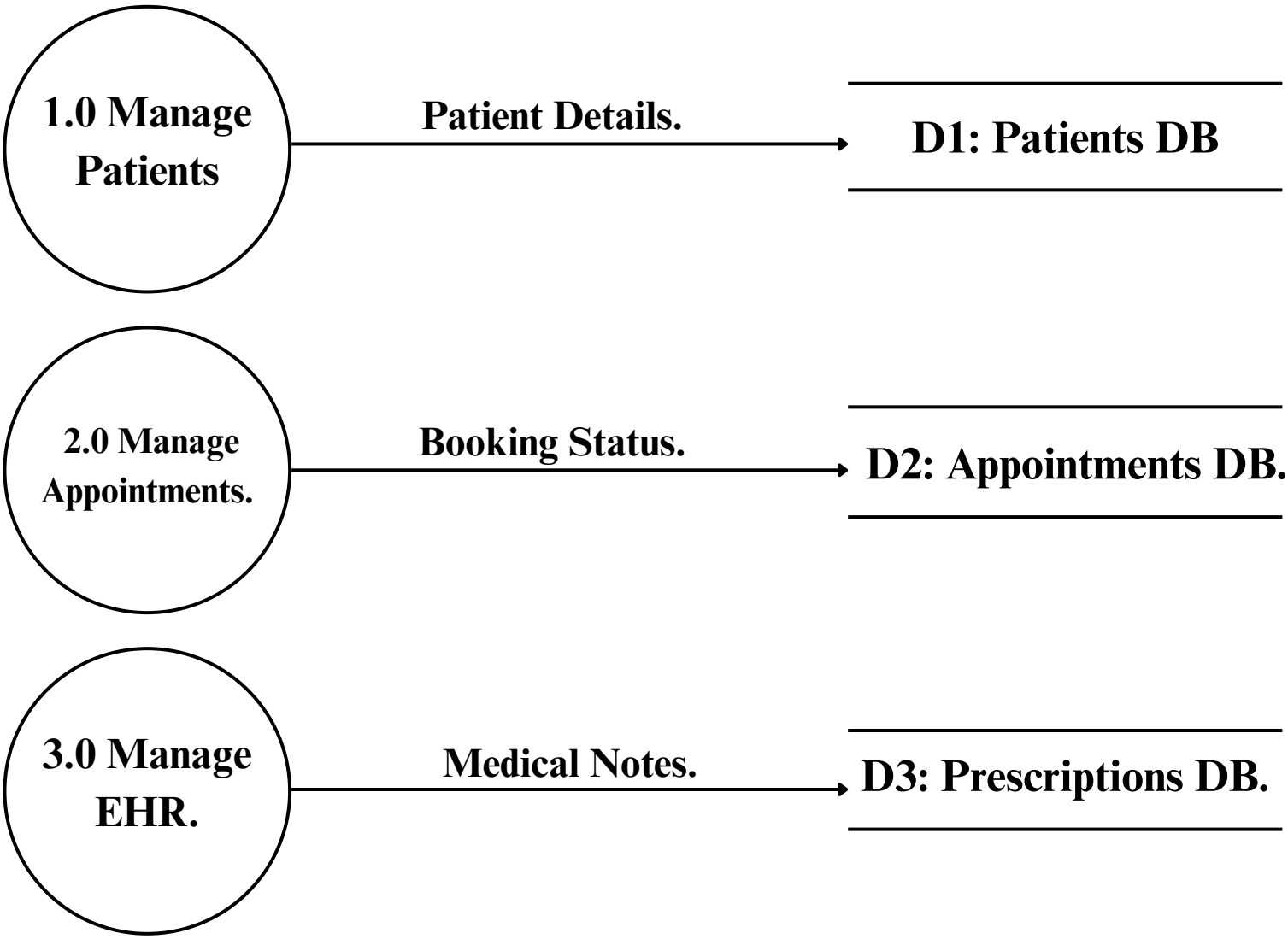


Dataflow Diagram (DFD Level 1)

Dataflow Diagram (DFD Level 1)

The Level 1 Data Flow Diagram breaks open the single "black box" established in the Context Diagram (Level 0) to reveal the internal mechanics of the CareCore system. It strictly maps how data from external entities routes through the core sub-processes and ultimately rests in specific data stores.

As a solo developer, mapping these internal pathways is crucial to ensure that data does not leak or route incorrectly before designing the backend database architecture. The diagram explicitly defines three primary processes (Managing Patients, Appointments, and EHRs) and their direct connections to three isolated database tables (D1, D2, and D3).



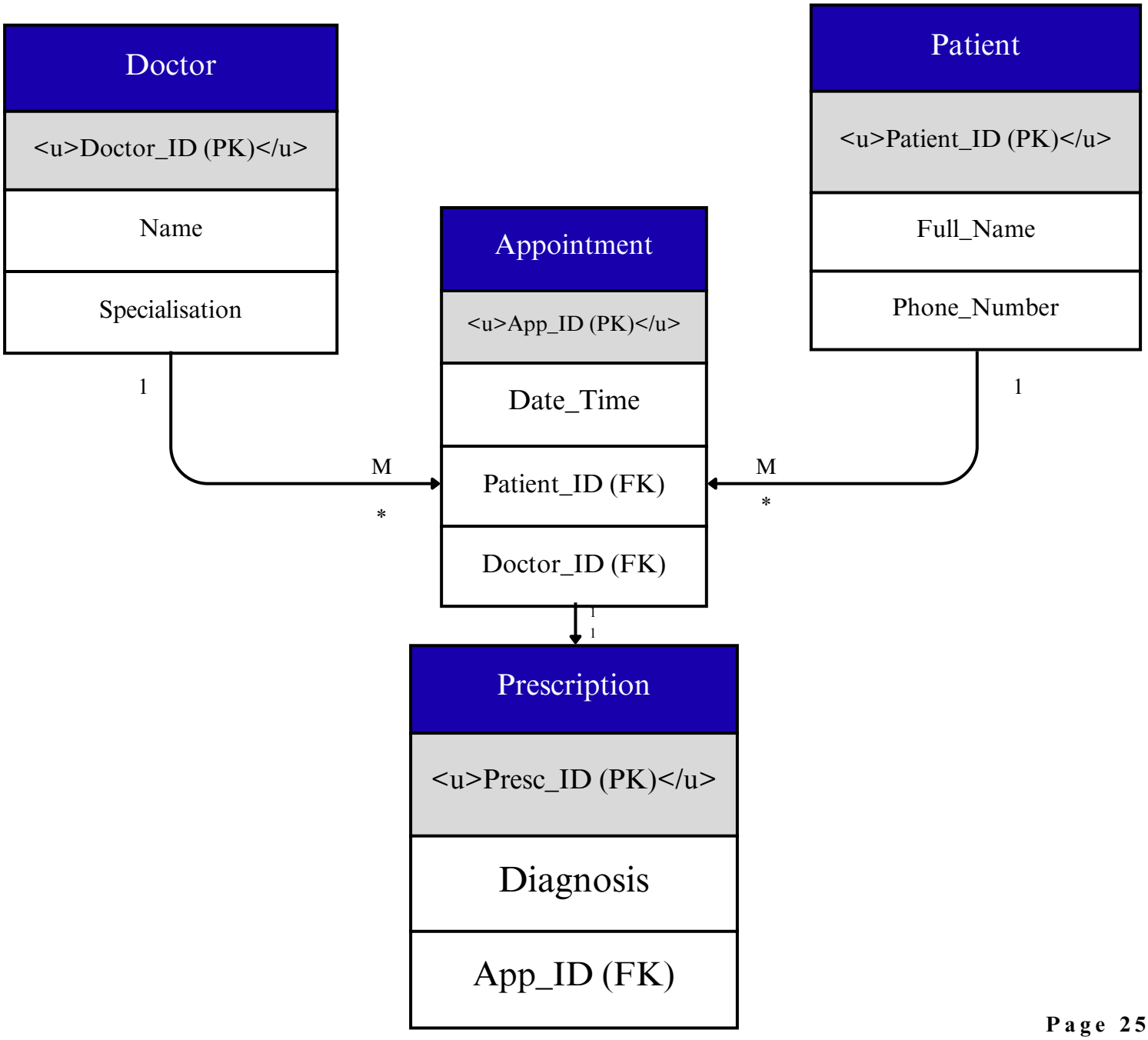
ERD Diagram (Database Design)

ERD Diagram (Database Design)

The Entity-Relationship Diagram (ERD) constitutes the exact physical database schema for the CareCore backend. Translating the data stores from the DFD Level 1, this model strictly defines the primary keys (PK), foreign keys (FK), and referential integrities required to eliminate data redundancy.

Architectural Relationships:

- One-to-Many (1:M): A single Patient can book multiple Appointments over time.
- One-to-Many (1:M): A single Doctor manages multiple Appointments daily.
- One-to-One (1:1): Every completed Appointment corresponds to exactly one secure Prescription record.



System Acquisition Strategy

When acquiring a new software solution, a crucial engineering and business decision must be made: whether to purchase Commercial Off-The-Shelf (COTS) software, outsource the project to a third-party vendor, or build a custom solution from scratch.

For the CareCore Smart Healthcare Management System, Custom In-House Development has been rigorously selected as the optimal acquisition strategy. Strategic Justifications:

- Academic Alignment: As a university project developed independently by Yousef Elnaggar, writing the code from scratch fulfills the core learning objectives of the Artificial Intelligence and Data Science programme.

- Total Architectural Control: Custom development grants 100% control over the database schema (ERD) and the strict data encryption standards (AES-256) defined in the nonfunctional requirements, which packaged software might not fully support.

- Cost Efficiency: It completely eliminates expensive vendor licensing fees and maintenance contracts, achieving the exact financial break-even point projected in the economic feasibility study (Month 4).

By adopting this strategy, the system remains highly tailored to the exact clinical problems identified during the requirement elicitation phase.

The Alternative Matrix (Decision Support)

The Alternative Matrix (Decision Support)

Having confirmed Custom In-House Development as the acquisition strategy, it is critical to select the optimal technology stack for building the cross-platform application. As a solo developer, the chosen framework must guarantee development speed without compromising the strict security and performance constraints (NFR-01 and NFR-02).

This Alternative Matrix mathematically evaluates two leading frameworks (Flutter and React Native) based on weighted engineering criteria to ensure a completely objective, data-driven decision.

Criteria	Weight (1-5)	Flutter (Score)	Flutter (Weighted)	React Native (Score)	React Native (Weighted)
Development Speed	4	5	**20**	4	**16**
System Performance	5	5	**25**	3	**15**
Data Security	5	4	**20**	4	**20**
Community Support	3	4	**12**	5	**15**
Total Score	-	-	**77**	-	**66**

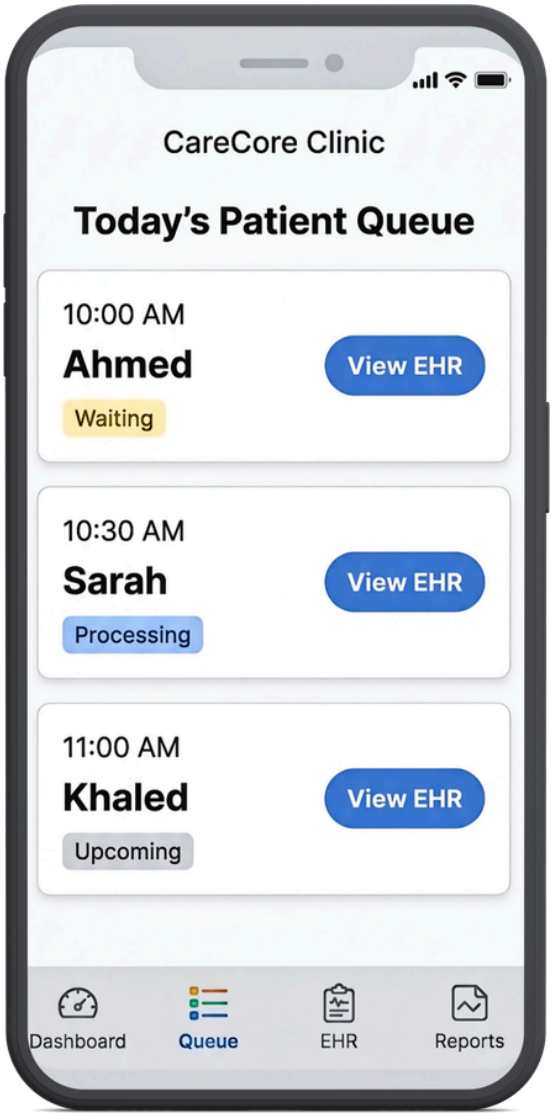
Conclusion: Based on the weighted mathematical calculation, Flutter achieved the highest total score (77 points) compared to React Native (66 points). Flutter's natively compiled rendering engine fully satisfies the 2.0-second performance requirement (NFR-01), making it the definitive choice for the CareCore project.

User Interface Design (Mobile App)

The mobile user interface is exclusively designed for the Patient actor. Driven by the usability constraint (NFR-05), the UI prioritises absolute simplicity and accessibility, ensuring that patients of all technical backgrounds can successfully navigate the system.

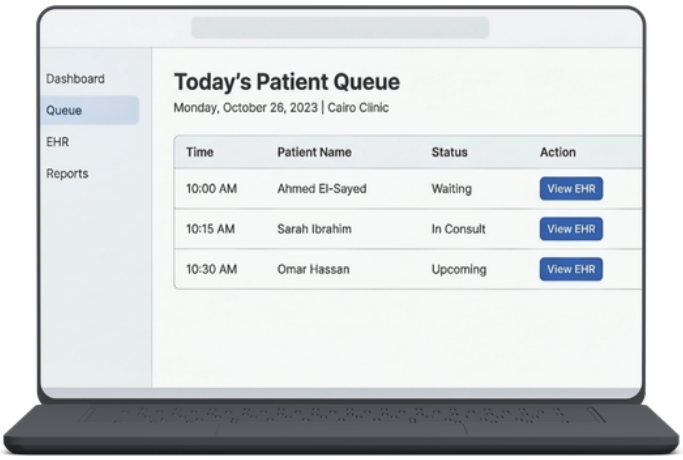
Key Design Features:

- Self-Service Dashboard: A clean, uncluttered home screen that immediately displays upcoming visits.
- Interactive Calendar: Allows the patient to visually select open time slots, fulfilling the core booking requirement (FR-02).
- One-Tap Booking: A prominent, high-contrast "Book Appointment" button ensures the entire booking process is completed in under three screen taps, drastically reducing the patient drop-off rate.



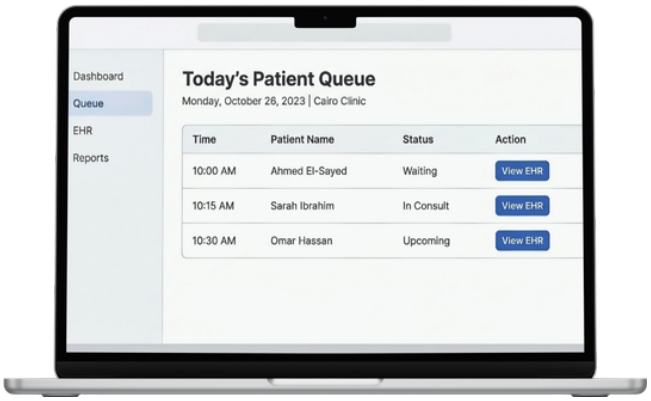
User Interface Design (Web Dashboard)

The Web Dashboard serves as the centralised control hub for the clinic's medical and administrative staff (Doctor and Secretary). Because medical clinics operate in high-pressure environments, the interface is designed to present maximum critical data with minimal visual clutter.



Key Design Features:

- Persistent Sidebar Navigation: Allows instant switching between the daily queue, patient records, and financial reports without losing context.



- Live Patient Queue: Provides the secretary with a real-time table of waiting patients, directly solving the waiting time complaints highlighted in the initial survey analysis (FR-04).

- Secure EHR Portal: Features an isolated, distraction-free digital medical form for the doctor to securely write and save digital prescriptions, fully satisfying requirement (FR-07).

Project Conclusion

The CareCore Smart Healthcare Management System represents a comprehensive, digitally transformative solution designed to eliminate the severe operational inefficiencies found in traditional, paper-based private clinics.

Through this documentation, the entire Software Development Life Cycle (SDLC) has been meticulously executed. By adhering to the strict sequential phases of the Waterfall methodology, the project successfully transitioned from initial requirement elicitation and rigorous feasibility analysis, into the creation of highly structured architectural blueprints. The meticulous formulation of UML diagrams, coupled with a robust Entity-Relationship Diagram (ERD), guarantees a secure, scalable, and highly cohesive backend infrastructure.

Ultimately, by leveraging mathematical decision-making through the Alternative Matrix, Flutter was selected for custom in-house development. As a solo software engineering effort by Yousef Elnaggar, this documentation proves that the CareCore system is completely prepared to transition from logical design into the physical implementation phase, promising to deliver a secure, GDPR-compliant, and highly efficient healthcare experience for both medical professionals and patients.

(Italic):
End of Documentation - CareCore System
Developed by: Yousef Elnaggar